

Problem

Find all pairs (a, b) of positive integers such that

$$ab \mid a^{2017} + b.$$

Source: HMMT 2017

§1 Separating the Product Divisibility

The condition

$$ab \mid a^{2017} + b$$

implies simultaneously

$$a \mid a^{2017} + b$$

and

$$b \mid a^{2017} + b.$$

The first divisibility immediately yields

$$a \mid b.$$

Therefore there exists a positive integer k such that

$$b = ak.$$

Substituting into the original condition gives

$$a^2k \mid a^{2017} + ak.$$

Dividing both sides by a ,

$$ak \mid a^{2016} + k.$$

This has exactly the same shape as the original problem, except that the exponent has decreased by one.

§2 Extracting the Hidden Descent

Suppose

$$ak \mid a^{2016} + k.$$

Reducing modulo a , we obtain

$$a \mid k.$$

Hence we may write

$$k = ak_1.$$

Substituting back,

$$a^2 k_1 \mid a^{2016} + a k_1.$$

Dividing by a again gives

$$a k_1 \mid a^{2015} + k_1.$$

Thus the same argument can be repeated.

Proceeding inductively, after 2016 repetitions we arrive at integers

$$\ell > 0$$

such that

$$a\ell \mid a + \ell.$$

Reducing modulo a once more yields

$$a \mid \ell.$$

Writing $\ell = at$, we obtain

$$a^2 t \mid a(1 + t),$$

and therefore

$$at \mid 1 + t.$$

Since at is a positive divisor of $1 + t$,

$$at \leq 1 + t.$$

Rearranging,

$$(a - 1)t \leq 1.$$

Because a, t are positive integers, the only possibilities are

$$t = 1, \quad a \in \{1, 2\}.$$

Thus

$$a = 1 \quad \text{or} \quad a = 2.$$

§3 Case 1: $a = 1$

The original condition becomes

$$b \mid 1 + b.$$

Hence

$$b \mid (1 + b) - b = 1.$$

Therefore

$$b = 1.$$

This yields the solution

$$(a, b) = (1, 1).$$

§4 Case 2: $a = 2$

The condition becomes

$$2b \mid 2^{2017} + b.$$

In particular,

$$b \mid 2^{2017}.$$

Hence b is a power of two:

$$b = 2^\omega, \quad 0 \leq \omega \leq 2017.$$

Substituting,

$$2^{\omega+1} \mid 2^{2017} + 2^\omega.$$

Factoring out 2^ω ,

$$2^{\omega+1} \mid 2^\omega (2^{2017-\omega} + 1).$$

Cancelling 2^ω gives

$$2 \mid 2^{2017-\omega} + 1.$$

The number $2^{2017-\omega}$ is even, so

$$2^{2017-\omega} + 1$$

is odd unless $2017 - \omega = 0$.

Therefore the divisibility is possible only when

$$\omega = 2017.$$

Hence

$$b = 2^{2017}.$$

This gives the solution

$$(a, b) = (2, 2^{2017}) .$$

§5 The Final Classification

The two cases above produce exactly the pairs

$$(1, 1) \quad \text{and} \quad (2, 2^{2017}) .$$

A direct verification shows that both satisfy

$$ab \mid a^{2017} + b .$$

Therefore

$$\boxed{(a, b) = (1, 1) \quad \text{or} \quad (a, b) = (2, 2^{2017}) .}$$